

18.06 Linear Algebra - Lectures 14, 15, 16
Textbook Notes - Sections 4.1, 4.2, 4.3

• Section 4.1

• Orthogonality of the four subspaces

Orthogonality of vectors, $x^T y = 0$, extends to subspaces:
 $V \perp W$

↳ when every vector in V is perpendicular to every vector in W

The four fundamental subspaces are ~~not~~ orthogonal in pairs and are orthogonal complements (dimensions add to full space, so each pair accounts for every direction).

The reason $CA^T \perp N(A)$:

↳ built into $Ax = 0$: each row dotted with x gives zero, so x is perpendicular to every row and the whole space as a result.

• Method: Finding the Orthogonal Complement of a Subspace V .

1. Spanning vectors of V are written as rows of a matrix M .
2. The complement $V^\perp$ is the Nullspace of M ($Mx = 0$).
3. Special Solutions form a basis for $V^\perp$.
4. The dimensions are checked: $\dim V + \dim V^\perp$ must equal n .

• Worked Example pg 196 [where $V = \text{span}\{(1, 3, 4), (5, 2, 7)\}$ in $\mathbb{R}^3$

$$M = \begin{bmatrix} 1 & 3 & 4 \\ 5 & 2 & 7 \end{bmatrix} \xrightarrow{A_2 - 5A_1} \begin{bmatrix} 1 & 3 & 4 \\ 0 & -13 & -13 \end{bmatrix}$$

Reduced 2nd row forces $x_2 = -x_3$, back-substitution into 1st row gives $x_1 = -x_3$.

→ Taking $X_3 = -1$ produces the basic vector $(1, 1, -1)$, so:

$$V_{\perp} = \text{span} \{ (1, 1, -1) \}$$

→ Result confirms $(1, 3, 4) \cdot (1, 1, -1) = 0$

$$(5, 2, 7) \cdot (1, 1, -1) = 0$$

and dimensions $\underline{2 + 1 = 3}$ fill the space.

→ As $V_{\perp}$ obtained from $N(M)$, the identity $N(V_{\perp}) = C(M^T)$ is the same computation.

Section 4.2

• Projections

→ When a vector b lies outside a subspace, the closest point within it, is the projection p .

→ p is the closest point as the error $\underline{e = b - p}$ is perpendicular to the subspace (all other options have a larger error by Pythagoras).

→ Construction formula

(1) Coefficient $\hat{x}$ is found.

(2) Next is projection p .

(3) Then the projection matrix P .

• Method: (onto a line, direction a)

The error $e = b - \hat{x}a$ is required to be perpendicular to a ,

and $\underline{a^T e = 0}$ yields the scalar directly:

$$1. \hat{x} = a^T b / a^T a = \frac{a^T b}{a^T a}$$

$$2. p = \hat{x}a$$

$$3. P = \frac{a \cdot a^T}{a^T a}$$

, a rank 1 matrix (a column times a row over a number).

4. Verify result via $a^T e = 0$.

• Worked Example pg 208-209

↳ projecting $b = (1, 1, 1)$ onto $a = (1, 2, 2)$

The scalar & projection are computed first:

$$\hat{x} = \frac{a^T b}{a^T a} = \frac{(1+2+2)}{(1+4+4)} = 5/9$$

$$p = (5/9) \cdot (1, 2, 2) = (5/9, 10/9, 10/9)$$

Error & orthogonality are checked

$$e = b - p = (4/9, -1/9, -1/9)$$

$$a^T e = 1 \cdot (4/9) + 2(-1/9) + 2(-1/9) = 0. \quad \checkmark$$

Projection matrix follows the same vectors

$$P = \frac{a a^T}{a^T a} = \left(\frac{1}{9} \right) \cdot \begin{bmatrix} 1 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{bmatrix}$$

Multiplying Pb reproduces p , and $P^2 = P$, holds, which expresses concretely that projecting an already-projected vector changes nothing.

• Method (onto a subspace, columns of a)

↳ Same perpendicularity, now stated as $A^T(b - A\hat{x}) = 0$, produces similar equations:

1. $A^T A$ and $A^T b$ are formed
2. The normal equations $A^T A \hat{x} = A^T b$ are solved for $\hat{x}$
3. $p = A \hat{x}$, and where the matrix is needed, $P = A(A^T A)^{-1} A^T$
4. Verify result via $A^T e = 0$.

The matrix $A^T A$ is symmetric and is invertible precisely when the columns of A are independent, which is the condition that guarantees a unique $\hat{x}$.

NB → {

- * $e \perp p$
- * $\|e\| = \sqrt{e^T \cdot e}$
- * $\hat{x} = p/a$

e in $N(A^T)$
 p in $C(A)$
 $\hat{x}$ in $C(A^T)$
 $N(A) \Rightarrow$

- Worked Example pg 211
↳ with $A = \begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix}$ and $b = \begin{pmatrix} 6 \\ 0 \end{pmatrix}$

→ A small symmetric system:

$$A^T A = \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix} \quad A^T b = \begin{bmatrix} 6 \\ 0 \end{bmatrix}$$

- The two rows read $C + D = 2$ & $3C + 5D = 0$
- Subtraction gives $D = -3$ and $C = 5$.

$$\hat{x} = \begin{pmatrix} 5 \\ -3 \end{pmatrix}$$

$$p = A\hat{x} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$$

$$e = b - p = \begin{pmatrix} 4 \\ 1 \end{pmatrix}$$

- The condition $A^T e = 0$ holds against both columns
- The solution $\hat{x} = \begin{pmatrix} 5 \\ -3 \end{pmatrix}$

◦ Section 4.3

• Least Squares Approximations

A least squares problem is a projection problem in different language. If $Ax = b$ has no solution (typically when there are more ~~solutions~~ equations than unknowns & b falls outside the column space), the quantity that can still be minimised is the squared error.

$$\text{Squared Error} \rightarrow \|b - Ax\|^2$$

Minimising is identical to projecting b onto the ~~column~~ columns, so the answer is again the normal equations.

The same destination is reached three ways

- ↳ by geometry (nearest point is the projection, e perpendicular to cd).
- ↳ by algebra (b splits as $p + e$, and $A\hat{x} = p$ is solvable)
- ↳ by calculus (setting $\frac{\partial E}{\partial x_i} = 0$)

↳ Only new skill is converting problem into matrix A and vector V .

• Method (best-fit line $b = C + Dt$)

1. Column one of A is all ones; ① } $A = \begin{bmatrix} ① & ② \end{bmatrix}$
column two ^{of A} holds the t -values; ② }
 b holds the measured values ③ } $b = \begin{bmatrix} ③ \end{bmatrix}$

↳ prev. pg 7
2. Either the normal equations are solved, or the 2×2 shortcut is filled in:
$$A^T A = \begin{bmatrix} m & \sum t_i \\ \sum t_i & \sum t_i^2 \end{bmatrix} \quad A^T b = \begin{bmatrix} \sum b_i \\ \sum t_i b_i \end{bmatrix}$$

3. The system is solved for $\hat{x} = \begin{pmatrix} C \\ D \end{pmatrix}$, giving line $b = C + Dt$

4. The fit minimises $E = \sum e_i^2$, the sum of square vertical distances between data & line.

• Worked Example pg 219-220

↳ fitting a line to $(0, 6)$; $(1, 0)$; $(2, 0)$

Setup produces $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}$ and $b = \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix}$

↳ same system is solved in Chapter 4.2.

Shortcuts are computed from the data.

$$m = 3, \quad \sum t_i = 3, \quad \sum t_i^2 = 5, \quad \sum b_i = 6, \quad \sum t_i b_i = 0$$
$$A^T A = \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix}; \quad A^T b = \begin{bmatrix} 6 \\ 0 \end{bmatrix}$$

↳ Since system matches projection e.g., solution: $\hat{x} = \begin{bmatrix} 5 \\ -3 \end{bmatrix}$ line: $b = 5 - 3t$

Fitted heights at $t = 0, 1, 2$ are $(5, 2, -1)$, against data $\begin{pmatrix} 6 \\ 0 \\ 0 \end{pmatrix}$ so

→ $(e = (1, -2, 1))$ and $E = 1 + 4 + 1 = 6.$

• Overview

Chapters 4.1 - 4.3 act as a single argument.

4.1 establishes that error must be $\perp$ to column space

4.2 converts that perpendicularity into $A^T A \hat{c} = A^T b$

4.3 ^aapplies real data to same equation.